

Ch 5: Arithmetic Progressions

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

Q1. Which of the following are APs? If they form an AP, find the common difference d and write three more terms: 2, 4, 8, 16, ... (PYQ)

Q2. Which of the following are APs? 0, -4 , -8 , -12 , ... Find d and write three more terms.

Q3. Which of the following are APs? $\sqrt{3}$, $\sqrt{6}$, $\sqrt{9}$, $\sqrt{12}$, ...

Q4. Find the common difference of the AP: $1/a$, $(3-a)/3a$, $(3-2a)/3a$ ($a \neq 0$). (PYQ 2019)

Q5. Find the common difference of the AP: -5 , -1 , 3 , 7 , ...

Q6. Find the 20th term of the AP: 2, 7, 12, ... (PYQ)

Q7. Find the 31st term of an AP whose 11th term is 38 and 16th term is 73. (PYQ 2019)

Q8. If the 3rd and 9th terms of an AP are 4 and -8 respectively, which term of the AP is zero? (PYQ 2020)

Q9. Find the 9th term from the end of the AP: 5, 9, 13, ..., 185. (PYQ 2016)

Q10. Find the n th term from the end of the AP: 3, 8, 13, ..., 253. (PYQ)

Q11. Which term of the AP: $-7, -12, -17, -22, \dots$ will be -82 ? Is -100 a term of this AP? (PYQ 2020)

Q12. Which term of the AP: 3, 15, 27, 39, ... will be 132 more than its 54th term? (PYQ)

Q13. Is 132 greater than the 54th term of the AP 3, 8, 13, ...? (PYQ)

Q14. Find how many two-digit numbers are divisible by 3. (PYQ 2019)

Q15. How many multiples of 4 lie between 10 and 250? (PYQ 2020)

Q16. How many terms of the AP: 45, 39, 33, ... must be taken so that their sum is 180? Explain the double answer. (PYQ 2019)

Q17. How many terms of the AP: 9, 17, 25, ... must be taken to get a sum of 636? (PYQ)

Q18. If the 7th term of an AP is $\frac{1}{9}$ and its 9th term is $\frac{1}{7}$, find its 63rd term. (PYQ 2020)

Q19. In an AP, if $d = -4$ and $a_7 = 4$, find the first term. (PYQ)

Q20. In an AP, if $a = 3.5$, $d = 0$, and $n = 101$, find a_n . (PYQ)

Q21. The first term of an AP is 5, the last term is 45, and the sum is 400. Find the number of terms and common difference. *(PYQ 2019)*

Q22. If $S_n = n^2p$ and $S_m = m^2p$ ($m \neq n$) for an AP, prove that $Sp = p^3$. *(PYQ HOTS)*

Q23. The n th term of an AP is given by $a_n = 3 + 4n$. Find the common difference and the first four terms. *(PYQ)*

Q24. The n th term of an AP is $a_n = 7 - 3n$. Find the common difference. Is this an increasing or decreasing AP? *(PYQ 2020)*

Q25. Find the sum of first 20 terms of an AP in which $a_3 = 7$ and $a_7 = 3$. *(PYQ 2022)*

Q26. Find the sum: $25 + 28 + 31 + \dots + 100$. *(PYQ)*

Q27. Find the sum of all odd numbers between 0 and 50. *(PYQ 2020)*

Q28. Find the sum of all even numbers from 1 to 250. *(PYQ)*

Q29. Find the sum of all integers from 1 to 100 that are multiples of 2 or 5. *(PYQ)*

Q30. The sum of the first n terms of an AP is $5n - n^2$. Find the n th term and the 20th term. *(PYQ 2019)*

Q31. The sum of the first n terms of an AP is $3n^2 + 4n$. Find the 25th term. *(PYQ 2020)*

Q32. If the sum of first n terms of an AP is $4n^2 + 2n$, find the n th term. *(PYQ)*

Q33. The 4th and 8th terms of an AP are 24 and 44 respectively. Find the sum of the first 10 terms. (PYQ 2023)

Q34. The sum of 4th and 8th terms of an AP is 24 and the sum of 6th and 10th terms is 44. Find the first three terms. (PYQ 2020)

Q35. Find the number of terms in the AP: 18, 15.5, 13, ..., -49.5 and find the sum of all its terms. (PYQ)

SECTION B - 3 MARKS

Q1. The 8th term of an AP is 31. If its 15th term exceeds its 11th term by 16, find the AP. (PYQ 2020)

Q2. The 8th term of an AP is equal to 3 times its 3rd term. If its 6th term is 22, find the AP. (PYQ)

Q3. The 9th term of an AP is equal to 6 times its 2nd term. If its 5th term is 22, find the AP. (PYQ)

Q4. The 19th term of an AP is equal to 3 times its 6th term. If its 9th term is 19, find the AP. (PYQ)

Q5. The 16th term of an AP is 1 more than twice its 8th term. If the 12th term is 47, find the n th term. (PYQ 2019)

Q6. If the 10th term of an AP is 47 and its first term is 2, find the sum of its first 15 terms. (PYQ 2020)

Q7. Find the sum of the first 40 positive integers divisible by 6. (PYQ 2019)

Q8. Find the sum of all multiples of 7 lying between 500 and 900. (PYQ)

Q9. Find the sum of all three-digit natural numbers which are multiples of 11. (PYQ 2019)

Q10. Find the sum of all natural numbers from 1 to 100 which are divisible by 4 or 5. (PYQ 2018)

Q11. Find the sum of all two-digit odd positive integers. (PYQ 2020)

Q12. Find the sum of integers between 100 and 200 that are divisible by 9. (PYQ 2020)

Q13. Find the sum of all natural numbers less than 100 which are divisible by 6. (PYQ)

Q14. If the sum of first 7 terms of an AP is 49 and the sum of first 17 terms is 289, find the sum of first n terms. (PYQ 2019 — *most repeated*)

Q15. If the sum of the first 14 terms of an AP is 1050 and its first term is 10, find the 20th term. (PYQ 2018)

Q16. The sum of the first 7 terms of an AP is 63 and the sum of its next 7 terms is 161. Find the 28th term. (PYQ)

Q17. The sum of the first seven terms of an AP is 182. If its 4th and 17th terms are in ratio 1:5, find the AP. (PYQ)

Q18. If the sum of n terms of an AP is $3n^2 + 5n$ and its m th term is 164, find the value of m . *(PYQ 2020)*

Q19. The sum of the first 25 terms of an AP whose n th term is given by $t_n = 2 - 3n$. *(PYQ)*

Q20. The n th term of an AP is given by $(-4n + 15)$. Find the sum of its first 20 terms. *(PYQ)*

Q21. If S_n denotes the sum of first n terms of an AP whose common difference is d and first term is a , find $S_n - 2S_{n-1} + S_{n-2}$. *(PYQ HOTS)*

Q22. The sum of 5th and 9th terms of an AP is 30. If its 25th term is three times its 8th term, find the AP. *(PYQ)*

Q23. The sum of 2nd and 7th terms of an AP is 30. If its 15th term is 1 less than twice the 8th term, find the AP. *(PYQ)*

Q24. If the sum of first m terms of an AP is same as the sum of its first n terms, show that the sum of its first $(m + n)$ terms is zero. *(PYQ 2019)*

Q25. The ratio of the sum of m and n terms of an AP is $m^2:n^2$. Show that the ratio of its m th and n th terms is $(2m - 1):(2n - 1)$. *(PYQ 2018)*

Q26. Show that the sum of an AP whose first term is a , second term b , and last term c is $(a + c)(b + c - 2a) / 2(b - a)$. *(PYQ HOTS)*

Q27. Find the sum of four consecutive numbers in an AP whose sum is 32 and whose product of extremes and middle terms are in ratio 7:15. (PYQ 2019)

Q28. The four terms in AP have sum 20 and sum of squares is 120. Find the terms. (PYQ)

Q29. Find three numbers in AP whose sum is 15 and product is 105. (PYQ)

Q30. The angles of a triangle are in AP. The greatest angle is twice the least. Find all angles. (PYQ 2019)

Q31. The angles of a polygon are in AP. The least angle is 120° and the common difference is 5° . Find the number of sides. (PYQ)

Q32. A contract on construction job specifies a penalty for delay of completion beyond a certain date as follows: Rs. 200 for the first day, Rs. 250 for the second day, Rs. 300 for the third day and so on. If the contractor delays by 30 days, how much does he pay as penalty? (PYQ 2020)

Q33. Subba Rao started work in 1995 at an annual salary of Rs. 5000 and received an increment of Rs. 200 each year. In which year did his income reach Rs. 7000? (PYQ)

Q34. A sum of Rs. 1600 is to be used to give 10 cash prizes to students. If each prize is Rs. 20 less than its preceding prize, find the value of each prize. (PYQ 2019)

Q35. In a flower bed, there are 23 rose plants in the first row, 21 in the second, 19 in the third and so on. There are 5 rose plants in the last row. How many rows are there in the flower bed? (PYQ 2020)

Q36. A spiral is made of successive semicircles with centres at alternate points A and B on a straight line with radii 0.5, 1.0, 1.5, 2.0, ... cm. What is the total length of the spiral of 13 consecutive semicircles? (PYQ 2019)

Q37. 200 logs are stacked in a manner: 20 logs in the bottom row, 19 in the next row, 18 in the next and so on. In how many rows are 200 logs placed and how many logs are in the top row? (PYQ 2018)

Q38. Ramkali saves Rs. 5 in the first week, Rs. 8 in the second, Rs. 11 in the third, and so on. In how many weeks will she save Rs. 20.75? (PYQ 2019)

Q39. A small terrace at a football ground comprises 15 steps, each of which is 50 m long and built of solid concrete. Each step has a rise of $\frac{1}{4}$ m and a tread of $\frac{1}{2}$ m. Find the total volume of concrete required to build the terrace. (PYQ)

Q40. How many terms of the AP: 24, 21, 18, ... must be taken so that their sum is 78? (PYQ)

Q41. Find the middle term of the AP: 6, 13, 20, ..., 216. (PYQ 2020)

Q42. Find the middle term of the AP: 213, 205, 197, ..., 37. (PYQ)

Q43. The digit in tens place of a two-digit number is three times the digit in units place. If the digits are reversed, the resulting number and the original number differ by 54. Find the AP formed by using the digits and verify if these digits form an AP. *(PYQ HOTS)*

SECTION C - 5 MARKS

Q1. If the sum of first 7 terms of an AP is 49 and that of first 17 terms is 289, find the sum of first n terms. *(PYQ 2019 — most repeated)*

Q2. The ratio of the sums of m and n terms of an AP is $m^2:n^2$. Show that the ratio of m th and n th terms is $(2m - 1):(2n - 1)$. *(PYQ 2018 — most repeated)*

Q3. If the sum of first n terms of an AP is $4n - n^2$, find the first term and the common difference. Hence find the sum of first 20 terms. Also find the n th term. *(PYQ 2020)*

Q4. In an AP, if the p th term is $1/q$ and q th term is $1/p$, prove that the sum of first pq terms is $(pq + 1)/2$. *(PYQ 2019)*

Q5. The sum of four consecutive numbers in AP is 32 and the ratio of the product of the first and last term to the product of two middle terms is 7:15. Find the numbers. *(PYQ 2019)*

Q6. Find the sum of all two-digit numbers which when divided by 4 yield 1 as remainder. (PYQ 2019)

Q7. Find the sum of all three-digit natural numbers that are multiples of 7. (PYQ)

Q8. Find the sum of all numbers between 250 and 1000 which are exactly divisible by 3. (PYQ 2018)

Q9. Find the sum of all multiples of 8 lying between 201 and 950. (PYQ 2020)

Q10. The sum of three numbers in AP is 12 and the sum of their cubes is 288. Find the numbers. (PYQ 2019)

Q11. Find three numbers in AP whose sum is 24 and whose sum of squares is 200. (PYQ)

Q12. The houses in a row are numbered consecutively from 1 to 49. Show that there exists a value of x such that the sum of numbers of houses preceding the house numbered x is equal to the sum of numbers of houses following x . Find x . (PYQ 2019 — *most repeated*)

Q13. The sum of the first n terms of an AP is given by $S_n = 3n^2 - 4n$. Find the AP and the 25th term. (PYQ 2020)

Q14. An AP consists of 50 terms of which 3rd term is 12 and the last term is 106. Find the 29th term. (PYQ 2019)

Q15. The 19th term of an AP is 52 and the 38th term is 128. Find the sum of its first 56 terms. (PYQ)

Q16. In an AP, if the 12th term is -13 and the sum of first 4 terms is 24, find the sum of first 10 terms. (PYQ 2020)

Q17. The sum of first 8 terms of an AP is 100 and the sum of first 19 terms is 551. Find the AP. (PYQ)

Q18. If the m th term of an AP is $1/n$ and the n th term is $1/m$, find the sum of first mn terms. (PYQ 2018)

Q19. A man repays a loan of Rs. 3250 by paying Rs. 20 in the first month and then increases the payment by Rs. 15 each month. How long will it take him to clear the loan? (PYQ 2020)

Q20. A sum of Rs. 700 is to be used to give seven cash prizes to students. If each prize is Rs. 20 less than its preceding prize, find the value of each prize. (PYQ 2019)

Q21. In a potato race, a bucket is placed at the starting point, which is 5 m from the first potato and the other potatoes are placed 3 m apart in a straight line. There are 10 potatoes in the line. A competitor starts from the bucket, picks up the nearest potato, runs back with it, drops it in the bucket, runs back to pick up the next potato, runs to the bucket, and so on. What is the total distance the competitor has to run? (PYQ 2019)

Q22. A ladder has rungs 25 cm apart. The rungs decrease uniformly in length from 45 cm at the bottom to 25 cm at the top. If the top and bottom rungs are 2.5 m apart, what is the length of the wood required for the rungs? (PYQ)

Q23. The digits of a positive number of three digits are in AP and their sum is 15. The number obtained by reversing the digits is 594 less than the original. Find the number. *(PYQ 2020)*

Q24. An AP has 50 terms. The 2nd term is 14 and the 50th term is 110. Find the sum of last 30 terms. *(PYQ 2019)*

Q25. Find the sum of last 10 terms of the AP: 8, 10, 12, ..., 126. *(PYQ 2020)*

Q26. If S_n denotes the sum of first n terms of an AP, prove that $S_{30} = 3(S_{20} - S_{10})$. *(PYQ 2019)*